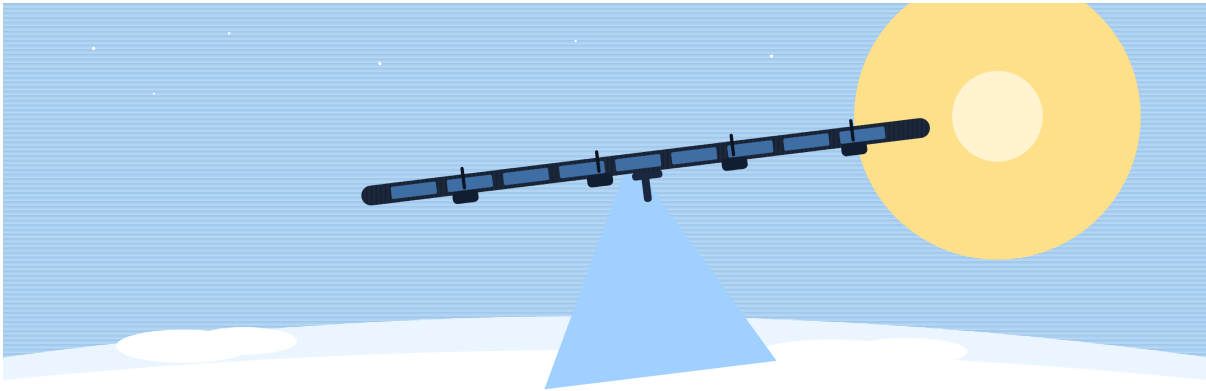


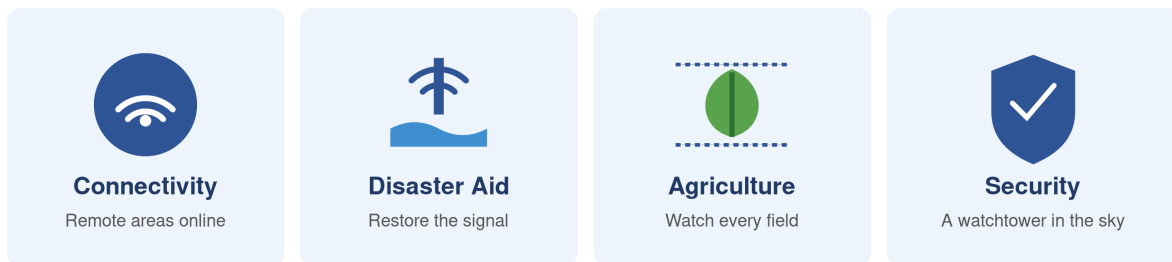


TECHNOLOGY · FEATURE

Getting to Know HAPS

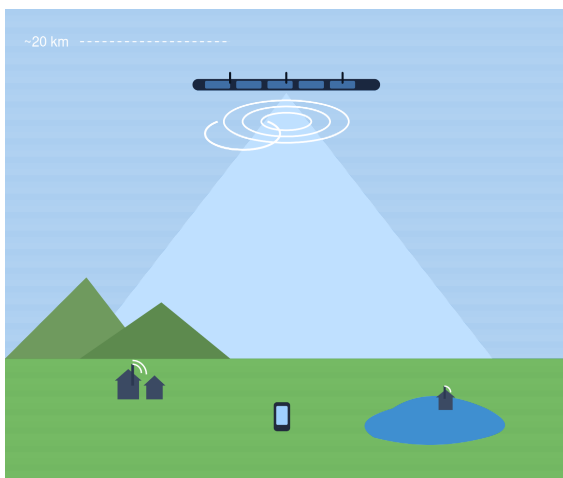
“Cell towers in the sky” that could change life in Thailand

Picture a solar-powered aircraft drifting almost motionless 20 kilometres overhead — twice as high as a passenger jet, above the clouds and all the weather, yet hundreds of times lower than a satellite. That, in essence, is a HAPS: a mobile-phone tower that hovers in the sky and, running on sunlight alone, can stay up there for months at a time.



Four ways HAPS can help — connectivity, disaster aid, agriculture, and security. (Original illustrations created for this article.)

What can HAPS do for us?



One HAPS covers a wide area — like a single tower lifted into the sky doing the work of dozens on the ground.

Bring coverage to remote areas. Mountain villages, islands out at sea, and

border regions that today have weak or no signal could make calls and use the internet on their existing phones — no special equipment needed.

Help during disasters. When a flood or earthquake knocks out ground towers, a HAPS can be sent up quickly to restore the signal over the disaster zone, so calls for help, search-and-rescue, and coordination happen without delay.

Watch over farmland. With high-resolution cameras and sensors, a HAPS keeps an eye on fields around the clock — flagging which areas are about to flood, which face drought, and where crops show signs of pests, so farmers can act in time.

Strengthen security. A HAPS acts like a watchtower in the sky, keeping continuous watch over borders and waters, and serving as a communications network the country can control on its own.

“Like lifting one tower into the sky that does the job of dozens on the ground — with almost no delay.”

Why this matters for Thailand

Thailand has mountains, islands, and border regions that signals struggle to reach. It is home

to around **12 million farmers** and faces floods and droughts almost every year — the floods of late 2024 alone caused tens of billions of baht in damage to farming. HAPS can address all of these at once: communications, agriculture, disaster response, and security.

Countries such as Japan and the United Kingdom, and global makers like Airbus and SoftBank, are already racing to launch real HAPS services. If Thailand develops the technology itself, the payoff is threefold — closing the access gap for remote communities, creating new jobs and industry at home, and building genuine technological self-reliance.

IN SHORT

- ▶ **HAPS = a cell tower in the sky** that floats 20 km up on solar power and stays aloft for months.
- ▶ **Four key benefits:** coverage in remote areas, signal recovery during disasters, support for agriculture, and national security.
- ▶ **A great fit for Thailand** — many remote areas, a large farming population, and frequent disasters.
- ▶ **Worth building ourselves** — to close the access gap, create local jobs, and gain technological self-reliance.

Note: A condensed, plain-language edition based on the full HAPS White Paper, drawing on the ITU, the HAPS Alliance, Airbus, and SoftBank. All illustrations are original works created for this article.